

①

$$\|u+v\|^2 = \langle u+v, u+v \rangle = \langle u, u \rangle + \langle v, u \rangle + \langle u, v \rangle + \langle v, v \rangle =$$

$$= \langle u, u \rangle + \langle v, v \rangle + \underbrace{\langle u, v \rangle + \overline{\langle u, v \rangle}}_{2\operatorname{Re}(\langle u, v \rangle)}$$

$$\left(a + \bar{a} = \operatorname{Re}(a) + i \cdot \operatorname{Im}(a) + \operatorname{Re}(a) - i \cdot \operatorname{Im}(a) = 2 \cdot \operatorname{Re}(a) \right)$$

$$\begin{aligned} \|u-v\|^2 &= \langle u-v, u-v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle = \\ &= \langle u, u \rangle + \langle v, v \rangle - 2\operatorname{Re}(\langle u, v \rangle) \end{aligned}$$

$$\|u+v\|^2 - \|u-v\|^2 = 4\operatorname{Re}(\langle u, v \rangle)$$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{\|u+v\|^2 - \|u-v\|^2}{4}$$

Využitím linearity
skalárního součinu.

②

$$\begin{aligned} \|u-iv\|^2 &= \langle u-iv, u-iv \rangle = \langle u, u \rangle + \langle -iv, u \rangle + \langle -iv, -iv \rangle + \langle u, -iv \rangle = \\ &= \langle u, u \rangle + \langle v, v \rangle + i\langle v, u \rangle - i\langle u, v \rangle \end{aligned}$$

$$\|u+iv\|^2 = \langle u+iv, u+iv \rangle = \langle u, u \rangle + i\langle u, v \rangle - i\langle v, u \rangle + \langle v, v \rangle$$

$$\|u-iv\|^2 - \|u+iv\|^2 = i\langle v, u \rangle - i\langle u, v \rangle - i\langle u, v \rangle + i\langle v, u \rangle =$$

$$= 2i(\langle v, u \rangle - \langle u, v \rangle) = 2i(\langle v, u \rangle - \overline{\langle v, u \rangle}) = 2i \cdot 2i \cdot \operatorname{Im}(\langle v, u \rangle) = 4 \cdot \operatorname{Im}(\langle u, v \rangle)$$

$$z = a + bi$$

$$z - \bar{z} = a + bi - (a - bi) = 2bi$$

$$\operatorname{Im}(\langle v, u \rangle) = -\operatorname{Im}(\langle u, v \rangle)$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$